AI-Powered Child Online Safety System

Child-focused cybersecurity, fraud prevention, conversational reporting and scam intelligence.

> Working project name: **Chinga**

CHALLENGE TRACK

Track 4 — AI for Child-Focused Cybersecurity and Online Fraud Prevention

VERSION

1.0 — Proposed Hackathon MVP

STATUS

Planning and Brainstorming

EVENT

5–6 November 2026

DOCUMENT GUIDE

Contents

Twenty sections covering the product vision, user journeys, AI security engine, platform channels, privacy, technical architecture and the hackathon demonstration.

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> **Scope note:** This document describes a proposed prototype. External integrations, providers, alert thresholds and the final product name remain subject to team decisions.

1. Project Overview

Chinga is an AI-powered cybersecurity system designed to protect children from online scams, phishing, fake websites, impersonation, malicious links, and online financial fraud.

The platform helps children check suspicious messages before trusting them and report online scams that have already happened.

The system also connects children to their parents or guardians, allowing parents to receive notifications when their children encounter serious cybersecurity threats.

In addition, the platform collects privacy-protected scam information and presents it through an intelligence dashboard. This dashboard helps authorised administrators, including potential UCC users, understand emerging scam patterns, identify organisations being impersonated, and investigate how scammers target children.

The platform will be accessible through a mobile application, Telegram bot, and SMS service, all connected to one central AI security engine.

2. Problem Statement

Children in Uganda are increasingly exposed to online fraud, including:

- Fake scholarship and educational opportunities.
- Fake Mobile Money rewards and promotions.
- Fraudulent websites impersonating trusted organisations.
- Phishing links designed to steal passwords, PINs, and personal information.
- Scammers promising gifts, money, or opportunities.
- Fraudulent messages received through WhatsApp, Telegram, SMS, and other platforms.

Many children cannot easily determine whether these messages are genuine.

Parents are also often unaware when their children encounter suspicious online activities.

At the same time, cybersecurity authorities may lack an accessible source of child-focused information about new scam campaigns, commonly impersonated organisations, and scam techniques.

3. Project Objectives

The system aims to:

1. Help children identify suspicious messages, links, and websites.
2. Give children simple, understandable cybersecurity advice.
3. Allow children to report scams through an AI conversation.
4. Notify parents or guardians when serious threats are detected.
5. Support children who lack mobile data through SMS.
6. Allow users to access the AI through Telegram.
7. Identify similar scam messages and emerging scam campaigns.

8. Provide administrators with anonymised statistics and relevant scam evidence.
9. Identify organisations frequently impersonated by scammers.
10. Protect children's privacy while collecting useful cybersecurity intelligence.

4. System Components

The solution will consist of five connected components.

Component	Purpose
Mobile Application	Child protection, parent registration, AI chat and notifications
Central AI Engine	Scam detection, analysis, verification, reporting and risk assessment
Telegram Bot	Access to the same AI through Telegram
SMS Integration	Scam checking and basic reporting without mobile internet
Web Admin Dashboard	Scam evidence, patterns, statistics and security intelligence

All components will communicate with the same backend and database.

5. User Roles

5.1 Child

Children can:

- Check suspicious messages and links.
- Upload screenshots for analysis.
- Chat with the AI assistant.
- Report scams that have already happened.
- Receive safety advice.
- Access the service through the app, Telegram, or SMS.
- Connect to a parent account using a secure code.

A child should not need an email address to use the service.

5.2 Parent or Guardian

Parents can:

- Register and verify an account.

- Add one or more children.
- Generate pairing codes for their children.
- Receive notifications about serious threats.
- View relevant safety alerts.
- Manage connected children and notification preferences.

Parents should not automatically access a child's complete private conversations.

5.3 Administrator

Authorised administrators can:

- View cyber-threat statistics.
- Examine relevant scammer messages.
- Identify scam patterns.
- See organisations being impersonated.
- Review suspected scam campaigns.
- Filter incidents by time, location, category, and risk level.
- Review reports and evidence according to their access permissions.

5.4 Guest User

A user without registration can access basic scam checking and safety advice.

Guest users will not receive family-linked notifications.

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6. Mobile Application Requirements

6.1 Welcome Screen

When the app opens for the first time, users should see simple options:

I'm a Parent

For parents who want to register and connect their children.

I Have a Family Code

For children whose parents have already created an account.

Check Something Now

For users who want basic safety advice without creating an account.

The interface must be simple, colourful, friendly, and appropriate for children.

6.2 Parent Registration

The proposed registration flow:

1. Parent enters their phone number.
2. Parent verifies the phone number using an OTP.
3. Parent creates a family account.
4. Parent adds a child's profile using a nickname and age bracket.
5. Parent chooses the notification settings.
6. The system generates a temporary pairing code.
7. The child enters that code on their device.
8. The backend links the child profile to the parent account.

The pairing code must be single-use, short-lived, and securely validated.

For the prototype, phone verification can be demonstrated using a controlled test flow. A production system will require proper guardian verification and additional safeguarding controls.

6.3 Child Home Screen

The child's home screen will have two main options.

Option A: "Is This Safe?"

This is for checking suspicious messages, links, and screenshots.

Option B: "Something Happened"

This is for children who have already experienced a scam or another cybersecurity incident.

The child should not be required to understand technical cybersecurity terms.

6.4 Intelligent Conversation Switching

The AI should recognise the child's actual intention.

For example, if a child selects "Is This Safe?" but says they have already entered their password on a suspicious website, the AI should move into the incident-support process.

Similarly, if a child selects "Something Happened" but only wants to check a suspicious link, the AI should switch to checking mode.

This switching must happen naturally without confusing the child.

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7. AI Scam Detection Engine

7.1 Supported Inputs

The AI security engine must accept:

- Plain text messages.

- Website URLs.
- Screenshots of suspicious messages.
- Suspicious messages forwarded or shared from other applications.
- Natural-language questions about possible scams.

For the hackathon MVP, image analysis will focus on screenshots.

7.2 Screenshot Analysis

A vision-capable LLM will analyse screenshots.

It should identify:

- Suspicious messages.
- Website links.
- Organisations being mentioned.
- Payment instructions.
- Requests for PINs, passwords, or personal information.
- Promised rewards or other suspicious offers.
- Messages that appear to come from a suspected scammer.

The system should extract relevant information and send it to the central security engine.

It should distinguish scammer messages from the child's messages where possible.

If the sender cannot be reliably identified, the application may ask the child to clarify.

7.3 Multi-Layer Scam Detection

The security engine will use several methods.

Layer 1: Pretrained Models

Use suitable phishing and spam-detection models to identify potentially fraudulent messages.

Layer 2: Threat Intelligence

Use available resources such as:

- PhishTank.
- URLhaus.
- OpenPhish.
- Other suitable cybersecurity databases.

Layer 3: Website Verification

The system should check available information such as:

- Domain registration age.
- Suspicious spelling and lookalike domains.
- Whether a URL matches an official organisation's website.
- Known malicious domain reports.

- Suspicious URL structures.

For example, a recently registered website pretending to represent a company established many years ago may be treated as suspicious.

A new domain alone must not automatically be classified as fraudulent.

Layer 4: LLM Reasoning

A capable LLM will examine the evidence, interpret the context, and explain the assessment.

The proposed AI provider may be Gemini, an OpenRouter-supported model, or another suitable model with the required capabilities.

The LLM should use verified information from external tools where available rather than assuming it can independently browse the internet.

7.4 Decision Process

The system must combine findings from the different security layers.

If the LLM and other detection systems disagree, the application should perform additional verification or return an uncertain result.

The LLM may provide the final natural-language explanation, but unsupported model claims must not override reliable technical evidence.

Possible outcomes:

- **Low Risk:** No clear signs of fraud were identified.
- **Suspicious:** Some warning signs were found.
- **High Risk:** Strong evidence suggests a scam.
- **Unable to Verify:** Insufficient evidence for a reliable conclusion.

The application must never guarantee that an unchecked or uncertain website is completely safe.

7.5 Child-Friendly AI Responses

AI responses must use simple English.

Example:

> “Be careful! This message looks suspicious. It is asking for your Mobile Money PIN. Real companies should not ask you to share your secret PIN. Do not enter it.”

The response should explain the reason for the warning and what the child should do next.

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8. AI-Powered Scam Reporting

The application must allow children to report incidents that have already happened.

8.1 Reporting Conversation

When a child selects “Something Happened,” the AI should begin a friendly conversation.

Example:

Child: Someone tricked me last week.

AI: I’m here to help. Can you tell me what happened?

Child: Someone said I won a scholarship and asked for money.

AI: Did you send them money?

Child: Yes, 20,000 shillings.

AI: Thank you for telling me. Do you have a message or screenshot you would like to share?

The AI must ask only relevant questions, avoid unnecessary repetition, and never request passwords or PINs.

8.2 Incident Information

Where available and appropriate, the system can capture:

- Incident type.
- Approximate date.
- Platform used.
- Suspected scammer messages.
- Suspicious website links.
- Organisation allegedly involved.
- Whether money was lost.
- Supporting evidence.
- General location, where appropriate.

8.3 Report Generation

The LLM will organise the supplied information into a clear report.

The child should be allowed to review the report.

Formal submission to UCC, Police, or another authority must use an approved reporting channel.

For the hackathon, a simulated report-submission workflow may demonstrate this functionality.

A confidential support route must remain available when involving a parent could put the child at risk.

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9. Parent Alert System

9.1 Purpose

The system should automatically notify connected parents when a child checks a serious cybersecurity threat.

9.2 Risk Levels

The prototype may use the following configurable risk-index thresholds:

Risk index	Action
0–39	Provide ordinary safety advice
40–79	Warn the child and explain the concern
80–89	Strong warning and notify the connected parent
90–100	Urgent warning and immediate parent-alert attempt

The risk index is an internal prioritisation score, not a scientifically calibrated probability of fraud.

Certain threats, such as clear requests for a Mobile Money PIN, may trigger urgent handling regardless of their numeric score.

9.3 Parent Notifications

Example notification:

> "Security Alert: Your child checked a suspicious message claiming to offer free MTN rewards. The message requested a Mobile Money PIN. The child has been warned not to proceed."

Notifications may be delivered through:

- Mobile push notifications.
- SMS fallback, where configured.

The backend should record notification delivery status and retry or provide appropriate failure handling.

Parents should receive relevant threat information without receiving their child's full private conversation.

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10. Telegram Bot Integration

The system will have a Telegram bot connected to the central backend.

Users can:

- Start a conversation with Chinga.
- Check suspicious messages.
- Submit links.
- Send screenshots.
- Ask cybersecurity questions.
- Report scams through conversation.

The Telegram bot should provide the same basic checking and reporting workflows as the mobile application.

Users with existing child profiles may connect their Telegram accounts using a secure one-time pairing process.

Telegram messages must be analysed through the same central AI engine.

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11. SMS Integration

The platform should support children without mobile data.

11.1 SMS Checking

A child sends a suspicious message to a designated SMS number.

The SMS provider forwards the message to Chinga's backend.

The AI analyses it and sends a short, understandable reply.

11.2 SMS Reporting

Users can also explain an incident through SMS.

The AI should ask a small number of questions and guide them through basic reporting.

11.3 Proposed Technology

Africa's Talking or another suitable SMS provider may be used.

For the hackathon, the SMS integration may use a sandbox or test number.

A toll-free SMS service through a future UCC or telecom partnership is a proposed expansion, not a confirmed existing service.

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12. Scam Wave Intelligence

12.1 Overview

Scam Wave Intelligence identifies similar suspicious messages and potential scam campaigns affecting multiple users.

It helps the system move beyond individual scam checks and identify wider cybersecurity patterns.

12.2 How It Works

When suspicious content is submitted, the engine extracts relevant threat information.

It compares that information against existing incidents using:

- Message similarity.
- Suspicious domain matches.
- Repeated payment instructions.

- Common impersonated organisations.
- Scam techniques.
- Submission times.
- General geographic patterns.

The engine should identify both exact duplicates and messages that use different wording but similar scam techniques.

12.3 Campaign Detection

Similar incidents can be grouped into suspected campaigns.

For example:

> "Possible fake Mobile Money reward campaign detected across several districts."

The system should separate suspected campaigns from confirmed campaigns.

Repeated submissions of the same message should be identified to avoid artificially increasing incident counts.

12.4 Early-Warning Intelligence

Administrators should be notified when the platform detects unusual patterns.

Possible examples:

- Sudden increase in fake scholarship messages.
- Multiple suspicious websites impersonating the same institution.
- Repeated Mobile Money PIN requests.
- Rapid growth in similar scam messages across several districts.

These are intelligence alerts, not automatic findings of criminal responsibility.

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13. UCC Administrator Dashboard

13.1 Dashboard Overview

The web-based administrator dashboard will provide an overview of cyber threats detected through the platform.

Suggested metrics:

- Total messages analysed.
- Suspicious messages detected.
- High-risk cases.
- Submitted incident reports.
- Emerging scam campaigns.
- Frequently used scam techniques.

- Most impersonated organisations.
- District-level incident trends.

The statistics must clearly represent submitted or checked incidents, not all scams occurring in Uganda.

13.2 Scam Evidence Viewer

The dashboard must include relevant scammer messages as evidence.

For example:

> "Congratulations! You have won UGX 500,000. Enter your Mobile Money PIN to receive your reward."

The dashboard should also display:

- Detected scam technique.
- Impersonated organisation.
- Relevant suspicious URLs, displayed safely.
- Risk assessment and supporting reasons.
- Similar submitted messages.
- Associated suspected campaign.

Children's names, private replies, personal identifiers, and unrelated conversations must be excluded from the general dashboard.

Sensitive evidence should only be accessible to authorised investigators through a restricted evidence viewer.

13.3 Impersonated Organisations

The dashboard must identify organisations frequently mentioned or impersonated in submitted scam messages.

Examples may include:

- Telecom operators.
- Banks and payment providers.
- Universities and schools.
- Government institutions.
- Scholarship organisations.
- Online businesses.

Important: Being impersonated does not mean an organisation is responsible for the scam.

Administrators should be able to examine trends and evidence and make their own conclusions.

13.4 Campaign Investigation

An authorised administrator should be able to open a suspected campaign and see:

- The campaign's observed message patterns.
- Related submitted incidents.

- Examples of scammer messages.
- Suspicious links.
- Impersonated organisations.
- Relevant district-level trends.
- First and latest observed submission times.
- Whether the campaign is suspected, under review, or confirmed.

The dashboard should support filtering, searching, and marking reviewed incidents.

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14. Privacy and Child Protection

Privacy must be built into the system from the beginning.

The following rules apply:

1. Collect only the information necessary for scam detection, parent notifications, and authorised reporting.
2. Do not automatically expose children's full conversations to administrators or parents.
3. Remove or mask sensitive personal details from scam evidence.
4. Use secure authentication and role-based permissions.
5. Protect submitted screenshots and incident reports.
6. Ensure formal reports are shared only through approved procedures.
7. Provide an appropriate confidential support path for sensitive incidents.
8. Never treat an LLM's risk score alone as sufficient proof of wrongdoing.
9. Use synthetic or simulated children's data for hackathon demonstrations.
10. Implement clear data retention and deletion rules.

The prototype must follow the applicable safeguarding requirements of the UCC hackathon.

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15. Proposed Technical Architecture

Mobile Application

Recommended: React Native with Expo.

Responsibilities:

- Parent registration.
- Child pairing.
- AI chat interface.
- Screenshot uploads.
- Sharing content to Chinga.

- Notifications.
- Account settings.

Web Dashboard

Recommended: Next.js with a suitable dashboard UI framework.

Responsibilities:

- Threat statistics.
- Incident evidence.
- Scam campaign monitoring.
- Organisation impersonation analysis.
- Report review.
- Administrator access control.

Backend

Recommended: Node.js with Express or NestJS.

Responsibilities:

- Authentication and parent-child linking.
- AI request handling.
- URL and threat checks.
- Risk assessment.
- Incident recording.
- Parent alerts.
- Telegram integration.
- SMS integration.
- Scam campaign analysis.
- Dashboard APIs.

Database

Recommended: PostgreSQL.

Proposed data entities:

- Parents.
- Child profiles.
- Connected communication channels.
- User sessions.
- Scam analysis records.
- Threat evidence.
- Incident reports.
- Scam campaigns.

- Parent alerts.
- Administrator accounts.
- Audit logs.

AI Services

A suitable LLM will handle:

- Natural-language conversations.
- Screenshot understanding.
- Scam explanations.
- Report conversations.
- Information extraction.
- Contextual threat assessment.

Additional phishing models, threat databases, and domain verification services will support technical detection.

The system will use one shared AI backend across all communication channels.

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16. Hackathon MVP Scope

Core Demonstration Requirements

The working prototype should demonstrate:

1. A parent registering and creating a child profile.
2. A child connecting through a pairing code.
3. A child checking a suspicious message or screenshot.
4. AI extracting evidence and identifying scam indicators.
5. The system verifying available URL and threat information.
6. A child receiving a clear safety explanation.
7. A serious threat triggering a parent notification.
8. The relevant scam evidence appearing in the admin dashboard.
9. Multiple similar messages being grouped into a suspected scam campaign.
10. The child using conversational scam reporting.

Additional Demonstration Goals

If integrations are ready, also demonstrate:

- Telegram chatbot interaction.
- SMS interaction through a test service.
- Telegram-to-parent account linking.
- Advanced dashboard filters and campaign comparisons.

Future Expansion

Potential future improvements include:

- Official WhatsApp Business integration.
- Toll-free SMS partnership.
- Additional local-language support.
- Partnerships with telecom operators and schools.
- Approved official reporting integrations.
- Advanced scam campaign forecasting.
- Wider deployment across Uganda and other African countries.

17. Testing and Acceptance Criteria

The MVP must be tested against clear expected outcomes.

Scenario	Expected result
Known phishing URL	System warns the child and explains the evidence
Legitimate official website	System avoids falsely labelling it fraudulent
Lookalike domain	System detects suspicious differences
Screenshot of a scam	AI extracts the relevant scammer message
Child chooses wrong option	AI switches to the appropriate workflow
High-risk threat	Parent alert is triggered if the account is linked and safeguards permit
Repeated scam message	System identifies existing evidence instead of blindly creating duplicate campaigns
Similar scam messages	System can group strongly related incidents
Telegram message	Same central AI engine processes the request
Private child information	General admin dashboard does not expose it
AI service unavailable	System returns a safe error and does not invent a verdict

The team must also evaluate detection accuracy using a labelled test set and report false positives and false negatives.

18. Proposed Live Demo Flow

During the hackathon presentation:

Scene 1 — Child Protection

A child receives a suspicious message promising a cash reward.

They submit the message through the mobile application.

The AI detects warning signs, checks supporting information, and explains why the message may be fraudulent.

Scene 2 — Parent Notification

The system identifies the incident as high-risk and sends an alert to the connected parent.

Scene 3 — Scam Intelligence

The admin dashboard receives a privacy-protected incident record showing the scammer's message, detected tactics, and impersonated organisation.

Scene 4 — Scam Wave Detection

Another similar message is submitted through Telegram or the application.

The system links the incidents and updates a suspected scam campaign.

Scene 5 — AI Reporting

A child tells the assistant that they were scammed previously.

The AI asks simple questions and prepares a structured report.

The demonstration will use simulated cases and clearly distinguish working features from proposed future integrations.

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19. Success Criteria

The prototype will be considered successful when:

- Children can submit suspicious content and receive useful safety advice.
- The AI can explain its findings in simple language.
- Screenshots can be analysed.
- Parent and child accounts can be securely connected.
- High-risk events can trigger appropriate parent notifications.
- The admin dashboard displays useful scam evidence and statistics.
- Similar incidents can be grouped into suspected scam campaigns.
- Children can explain past scams through an AI conversation.
- Relevant personal information is protected.
- The team demonstrates working integration with the UCC Testbed infrastructure or provisioned services and reports its evaluation results.

20. Project Vision

The long-term vision is to create a child-focused cybersecurity protection network that works across mobile applications, messaging platforms, and SMS.

The platform should help children make safer decisions online, allow parents to respond to serious cybersecurity threats, and provide authorised institutions with useful information about scam techniques and emerging fraud campaigns.

The aim is to move from protecting one child against one scam to using privacy-protected intelligence to help protect many children.

Core Value Proposition:

> "An AI-powered cybersecurity companion that helps children recognise scams, seek help when something goes wrong, keeps parents informed about serious threats, and helps authorities identify emerging scam campaigns before more children become victims."

End of PRD — Version 1.0

Project name, exact technology providers, alert thresholds, and optional integrations remain subject to final team decisions.